**Practical No. 1**

**Aim: Create different blinking patterns with LEDs by connecting to GPIO pins of Raspberry Pi.**

**Connections:**

1. **Connect VGA to HDMI convertor, with Raspberry Pi & monitor.**
2. **Connect the USB keyboard and mouse with Raspberry Pi.**
3. **Connect the micro USB 5v Power supply to Raspberry Pi.**
4. **Connect Raspberry Pi to breadboard using jumper wires.**
5. **Connect the cathode of LED to the Ground pins. Physical pin 13 to led pin 4**
6. **Connect anode of the LED with resistors. Physical pin 6 to led ground**

**Connect the anode of the LED with the 220 Ω resistor & connect the resistor to 3.3 power i.e. Pin number 1 of GPIO header.**

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)

# Set up GPIO pins

GPIO.setup(17, GPIO.OUT)
GPIO.setup(18, GPIO.OUT)
GPIO.setup(27, GPIO.OUT)
GPIO.setup(22, GPIO.OUT)

try:
    while True:

        # Pattern 1: All LEDs ON
        print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")
        GPIO.output(17, GPIO.HIGH)
        GPIO.output(18, GPIO.HIGH)
        GPIO.output(27, GPIO.HIGH)
        GPIO.output(22, GPIO.HIGH)
        print("Pattern 1: All LEDs ON")
        time.sleep(0.4)

        # Pattern 2: All LEDs OFF
        GPIO.output(17, GPIO.LOW)
        GPIO.output(18, GPIO.LOW)
        GPIO.output(27, GPIO.LOW)
        GPIO.output(22, GPIO.LOW)
        print("Pattern 2: All LEDs OFF")
        time.sleep(0.4)

        # Pattern 3: LEDs 17 and 27 ON, 18 and 22 OFF
        GPIO.output(17, GPIO.HIGH)
        GPIO.output(18, GPIO.LOW)
        GPIO.output(27, GPIO.HIGH)
```

```
        GPIO.output(22, GPIO.LOW)
        print("Pattern 3: LEDs 17 and 27 ON")
        time.sleep(0.4)

        # Pattern 4: LEDs 17 and 27 OFF, 18 and 22 ON
        GPIO.output(17, GPIO.LOW)
        GPIO.output(18, GPIO.HIGH)
        GPIO.output(27, GPIO.LOW)
        GPIO.output(22, GPIO.HIGH)
        print("Pattern 4: LEDs 18 and 22 ON")
        time.sleep(0.4)

except KeyboardInterrupt:

    print("Program stopped")

finally:
    GPIO.cleanup()
```

**Practical No. 3**

**Aim: Interface TM1637 with Raspberry Pi and Display current time.**

**Connections:**

1. **Connect VGA to HDMI convertor, with Raspberry Pi & monitor.**
2. **Connect the USB keyboard and mouse with Raspberry Pi.**
3. **Connect the micro USB 5v Power supply to Raspberry Pi.**
4. **Connect tm1637 to Raspberry Pi using jumper wires.**

| TM1637 Pin | Raspberry Pi Pin |
|---|---|
| CLK | 23rd Pin (GPIO 11) |
| DIO | 24th Pin (GPIO 08) |
| VCC | 2nd Pin |
| GND | 6th Pin |

**Program Code:**

```
import sys
import time
import datetime
import RPi.GPIO as GPIO
import tm1637

# Names
print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")

# Initialize 7-segment display
Display = tm1637.TM1637(11, 8, tm1637.BRIGHT_TYPICAL)

Display.Clear()
Display.SetBrightness(1)

try:
    while True:
        now = datetime.datetime.now()

        hour = now.hour
        minute = now.minute
        second = now.second

        currenttime = [
            int(hour / 10),
            hour % 10,
            int(minute / 10),
            minute % 10
        ]
        Display.Show(currenttime)
        Display.ShowDoublepoint(second % 2)

        print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")
        print("Time:", now.strftime("%H:%M:%S"))

        time.sleep(1)
```

```
except KeyboardInterrupt:
    print("Program stopped")

finally:
    Display.Clear()
    GPIO.cleanup()
```

**Practical No. 4**

**Aim: Interface GPS module with Raspberry Pi.**

**Connections:**

1. **Connect VGA to HDMI convertor, with Raspberry Pi & monitor.**
2. **Connect the USB keyboard and mouse with Raspberry Pi.**
3. **Connect the micro USB 5v Power supply to Raspberry Pi.**
4. **Jumper wires to connect GPS module with Raspberry PI**

- **Vcc to Pin 2**
- **Tx to Pin 10 (Rx)**
- **Gnd to Pin 6**

**This module provides nonstop NMEA (National Marine Electronics Association) data strings to the TX pin resulting in GPS information.**

**Terminal Commands:**

**Save → gpss.py**

**Run → terminal →**

```
sudo python gpss.py
```

**Program Code:**

```
import serial
import time
import string
import pynmea2

print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")
print("----------------------------------------")

port = "/dev/serial0"
ser = serial.Serial(port, baudrate=9600, timeout=0.5)

try:
    while True:
        newdata = ser.readline().decode(
            "ascii", errors="replace"
        ).strip()

        if newdata.startswith("$GPRMC"):
            newmsg = pynmea2.parse(newdata)
            lat = newmsg.latitude
            lng = newmsg.longitude
            gps = "Latitude=" + str(lat) + " and Longitude=" + str(lng)
            print(gps)

except KeyboardInterrupt:
    print("Program stopped")
```

```
finally:
    ser.close()
```

**Practical No. 5**

**Aim: Interface ADS 1115 with Raspberry Pi to act as oscilloscope.**

**Connections:**

1. **Connect VGA to HDMI convertor, with Raspberry Pi & monitor.**
2. **Connect the USB keyboard and mouse with Raspberry Pi.**
3. **Connect the micro USB 5v Power supply to Raspberry Pi.**
4. **Connect ADC to Raspberry Pi using jumper wires.**

| ADS1115 Pin | Raspberry Pi Pin |
|---|---|
| VDD | 17th Pin (3.3 V) |
| GND | 9th Pin (GND) |
| SCL | 5th Pin (GPIO 03) |
| SDA | 3rd Pin (GPIO 02) |

**Terminal Commands:**

**Step 1: Enable Raspberry Pi I2C interface. To enable the I2C, from the terminal, run:**

```
sudo raspi-config
```

**When the configuration panels open, select interface options, select I2C and click enable.**

**Step 2: Update the Raspberry pi**

```
sudo apt-get update
sudo apt-get upgrade
```

**Step 3: Install the Adafruit ADS1115 library for ADC**

```
cd ~
```

**then install the build-essentials by running; (Python or Python3)**

```
sudo apt-get install build-essential python-dev python-smbus git
```

**Next, clone the Adafruit git folder for the library by running;**

```
git clone [UNCLEAR IN SOURCE – VERIFY] (URL hidden by the on-screen toolbar
in the slide photo; only the start,
"https://github.com/adafruit/Adafruit_Python_…", is partly visible)
```

**Step 4: Test the library and I2C communication. While still in the Adafruit_Python_ADS1x15 folder, change directory to the examples directory by running;**

```
cd examples
```

**Run the example using:**

```
python simpletest.py
python3 simpletest.py
```

**Step 5: Install Matplotlib Python or Python3**

```
sudo apt-get install python-matplotlib
```

**Program Code:**

```
import time
import numpy.core.multiarray
import matplotlib.pyplot as plt
import Adafruit_ADS1x15
import RPi.GPIO as GPIO

adc = Adafruit_ADS1x15.ADS1115()

GAIN = 16
val = []
cnt = 0

print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")

plt.ion()
adc.start_adc(0, gain=GAIN)

print("Reading ADS1115 channel 0")

def makeFig():
    plt.ylim(-5000, 5000)
    plt.title("Oscilloscope")
    plt.grid(True)
    plt.ylabel("ADC outputs")
    plt.plot(val, 'r-o', label='lux')
    plt.legend(loc='lower right')

try:
    while True:
        value = adc.get_last_result()
        print("Channel0:{}".format(value))
        time.sleep(0.5)

        val.append(int(value))
        plt.clf()
        makeFig()
        plt.pause(0.00001)

        cnt += 1
        if cnt > 50:
            val.pop(0)
            cnt = cnt - 1

except KeyboardInterrupt:
    print("Program stopped")
```

```
    plt.close()
```

**Practical No. 6**

**Aim: Interface 16x2 LCD with Raspberry Pi using I2C module to display text.**

**Connections:**

**Terminal Commands:**

**Step 2: Install I2C Tools. After Raspberry Pi restarts, open the Terminal and install the I2C tools:**

```
sudo apt update
sudo apt install i2c-tools
```

**Step 3: Check the I2C Address. Run:**

```
sudo i2cdetect -y 1
```

**Step 4: Install the Required Python Library. For a simple setup, install the required packages:**

```
sudo apt install python3-rplcd python3-smbus
```

**If python3-rplcd is not available on your Raspberry Pi OS version, use a Python virtual environment:**

```
python3 -m venv lcd-env
```

**Activate it:**

```
source lcd-env/bin/activate
```

**Then install:**

```
pip install RPLCD smbus2
```

**Program Code:**

```
import time
import board
import digitalio
import adafruit_character_lcd.character_lcd as characterlcd

lcd_columns = 16
lcd_lines = 2

lcd_rs = digitalio.DigitalInOut(board.D26)

lcd_en = digitalio.DigitalInOut(board.D19)

lcd_d4 = digitalio.DigitalInOut(board.D25)

lcd_d5 = digitalio.DigitalInOut(board.D24)
```

```
lcd_d6 = digitalio.DigitalInOut(board.D22)

lcd_d7 = digitalio.DigitalInOut(board.D27)

lcd_backlight = digitalio.DigitalInOut(board.D4)

lcd = characterlcd.Character_LCD_Mono(

    lcd_rs, lcd_en, lcd_d4, lcd_d5, lcd_d6, lcd_d7,

    lcd_columns, lcd_lines, lcd_backlight

)

lcd.message = "Hello World"

time.sleep(5)
lcd.clear()
```

## Practical No. 7

**Aim: Interface RFID module with Raspberry Pi to read and write to RFID tags/cards.**

**Connections:**

1. **Connect VGA to HDMI convertor, with Raspberry Pi & monitor.**
2. **Connect the USB keyboard and mouse with Raspberry Pi.**
3. **Connect the micro USB 5v Power supply to Raspberry Pi.**
4. **Connect RFID RC522 to Raspberry Pi using jumper wires.**

| RFID RC522 Pin | Raspberry Pi Pin |
|---|---|
| SDA | Pin 24 |
| SCK | Pin 23 |
| MOSI | Pin 19 |
| MISO | Pin 21 |
| GND | Pin 6 |
| RST | Pin 22 |
| 3.3v | Pin 1 |

**Terminal Commands:**

```
pip3 install mfrc522
```

**OR**

```
pip3 install mfrc522 –break-system-packages
```

**Enable SPI on the Raspberry Pi:**

```
sudo raspi-config → Interface Options → SPI → [UNCLEAR IN SOURCE – VERIFY]
(rest of line hidden by toolbar in slide image)
```

**Program Code:**

**Write Code:**

```
import RPi.GPIO as GPIO
from mfrc522 import SimpleMFRC522

rfid = SimpleMFRC522()

print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")

try:
    print("Hold the RFID tag near the reader...")
    rfid.write("Milan")
    print("Data written successfully!")

except KeyboardInterrupt:
    print("Program stopped")
```

```
finally:
    GPIO.cleanup()
```

**Read Code:**

```
import RPi.GPIO as GPIO
from mfrc522 import SimpleMFRC522


rfid = SimpleMFRC522()

print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")

try:
    while True:

        id, text = rfid.read()

        print(id)
        print(text)
except KeyboardInterrupt:

    print("Program stopped")


finally:
    GPIO.cleanup()
```

## Practical No. 9

**Aim: Interface Pi Camera with Raspberry Pi to capture image and video.**

**Connections:**

1. **Connect VGA to HDMI convertor, with Raspberry Pi & monitor.**
2. **Connect the USB keyboard and mouse with Raspberry Pi.**
3. **Connect the micro USB 5v Power supply to Raspberry Pi.**
4. **Pi Camera Module.**

- **Open the Raspberry Pi camera notch**
- **Insert camera module in the notch**
- **Close the notch.**

**Terminal Commands:**

```
sudo raspi-config
sudo apt --fix-broken install
sudo apt update
sudo apt upgrade
sudo apt-get install python-picamera
```

**or**

```
sudo apt-get install python3-picamera2 -y
```

```
cd Desktop
mkdir camera
cd camera
raspistill -o image.jpg
```

**OR**

```
rpicam-still -o image.jpg
```

```
ls
raspistill -o movie.h264 -t 500
```

**Program Code:**

```
from picamera import PiCamera
from time import sleep

print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")

camera = PiCamera()

camera.resolution = (1280, 780)

camera.start_preview()
camera.start_recording('video.h264')
```

```
for i in range(5):
    sleep(5)
    camera.capture('images%s.jpg' % i)

camera.stop_recording()
camera.stop_preview()
camera.close()
```

## Practical No. 18

**Aim: Write a program to read the CPU%, Memory% and CPU temperature of Raspberry Pi.**

**Connections:**

**Program Code:**

```
import psutil
import time

print("Ayesha, Saamiya, Mantasha, Harsh, Piyush")

while True:
    cpu = psutil.cpu_percent(interval=1)
    memory = psutil.virtual_memory().percent

    with open("/sys/class/thermal/thermal_zone0/temp", "r") as file:
        temp = int(file.read()) / 1000

    print("CPU Usage:", cpu, "%")
    print("Memory Usage:", memory, "%")
    print("CPU Temperature:", temp, "°C")
    print("----------------------")

    time.sleep(2)
```

**Practical No. 20**

**Aim: Installing Windows 10 IoT Core on Raspberry Pi.**

**Connections:**

**Procedure (from teacher's slides):**

1. **Go to the official website of Microsoft which is https://docs.microsoft.com/en-us/windows/iot-core/downloads Click on the option "Download the Windows 10 IoT Core Dashboard".**
2. **After downloading the above image, click the "Install" option.**
3. **After installation click on "Run" option.**
4. **IoT Dashboard window will appear. Click on "Set up a new device" option.**
5. **A window will appear as shown below. Insert the SD card & select the drive. Set the root password and check the software license terms checkbox. Finally click on "Download & Install" option.**
6. **A pop-up window will appear to erase the SD Card. Click on "Continue" option to erase existing data if any.**
7. **This will start downloading Windows 10 IoT Core. Once downloaded it will start with flashing the SD card/Device.**
8. **Lastly a popup menu will appear to format or cancel the SD card. Simply click on "Cancel Option".**

**practical 20: install the OS on Raspberry Pi and run the demo program**

**part A: prepare the SD card**

1. **on a PC, open https://www.raspberrypi.com/software/ and download Raspberry Pi Imager, then install it.**
2. **insert a microSD card (8 GB or more) into the PC.**
3. **open Raspberry Pi Imager.**
4. **click "Choose Device" and select your Pi model.**
5. **click "Choose OS" and select Raspberry Pi OS (64-bit). use the 32-bit version if your Pi model needs it.**
6. **click "Choose Storage" and select your SD card. check that it's the right drive, because it gets erased.**
7. **click "Next". when it asks about OS customization, set a hostname, username and password, and turn on SSH if you want it. use "Edit Settings" for this.**
8. **confirm the warning that the card will be erased. the Imager writes and verifies the OS. when it says "write successful", remove the card.**

**part B: boot the Pi**

**9. put the SD card into the Raspberry Pi.**

**10. connect the HDMI monitor, keyboard and mouse, then plug in the power supply.**

**11. the Pi boots, which takes a minute or two the first time. follow the welcome setup for country, language, password and Wi-Fi, and let it check for updates.**

**12. the Raspberry Pi desktop appears, which means the OS is installed.**

**part C: run the demo program**

**13. open Terminal and update the system:**

```
sudo apt update
sudo apt upgrade
```

14. **open Thonny (Menu → Programming → Thonny Python IDE) and type a simple program:**

```
print("Hello from Raspberry Pi")
```

15. **save it as demo.py and press Run. the output appears in the shell panel.**
16. **you can also run it from the terminal with python3 demo.py.**